

Supply Chain OTIF & Bottle Neck Analytics

1. Project Overview

This project evaluates global supply chain fulfillment performance using transactional logistics data from **58,136 unique order line items** across global markets.

The primary objective is to measure **On-Time In-Full (OTIF)** compliance, isolate carrier Service Level Agreement (SLA) failures, quantify regional revenue at risk, and identify operational bottlenecks across dispatch and delivery channels. By connecting SQL database querying in Supabase, exploratory grid modeling in Quadratic, and Python visual analytics in Google Colab, this report provides data-driven strategic recommendations to optimize logistics velocity, improve customer satisfaction, and protect working capital.

2. Dataset Summary

Total Transactions (Rows): 58,136 clean order line records.

Attributes (Columns): 12 core supply chain and financial features including order_id, order_item_id, customer_id, order_region, order_country, category_name, shipping_mode, days_for_shipping_real, days_for_shipment_scheduled, sales, and order_profit_per_order.

Missing & Null Data Handling: Verified dataset completeness across core logistics parameters. Cleaned order status discrepancies and excluded invalid records to ensure high data integrity.

Core Feature Engineering:

- is_on_time: Binary flag (1 if actual transit days $\leq$ scheduled transit days, else 0).
- is_in_full: Binary flag (1 if order was completed without cancellation or fraud, else 0).
- is_otif: Binary flag (1 if both is_on_time = 1 AND is_in_full = 1, else 0).
- delay_variance_days: Difference between actual days shipped and scheduled shipment days.

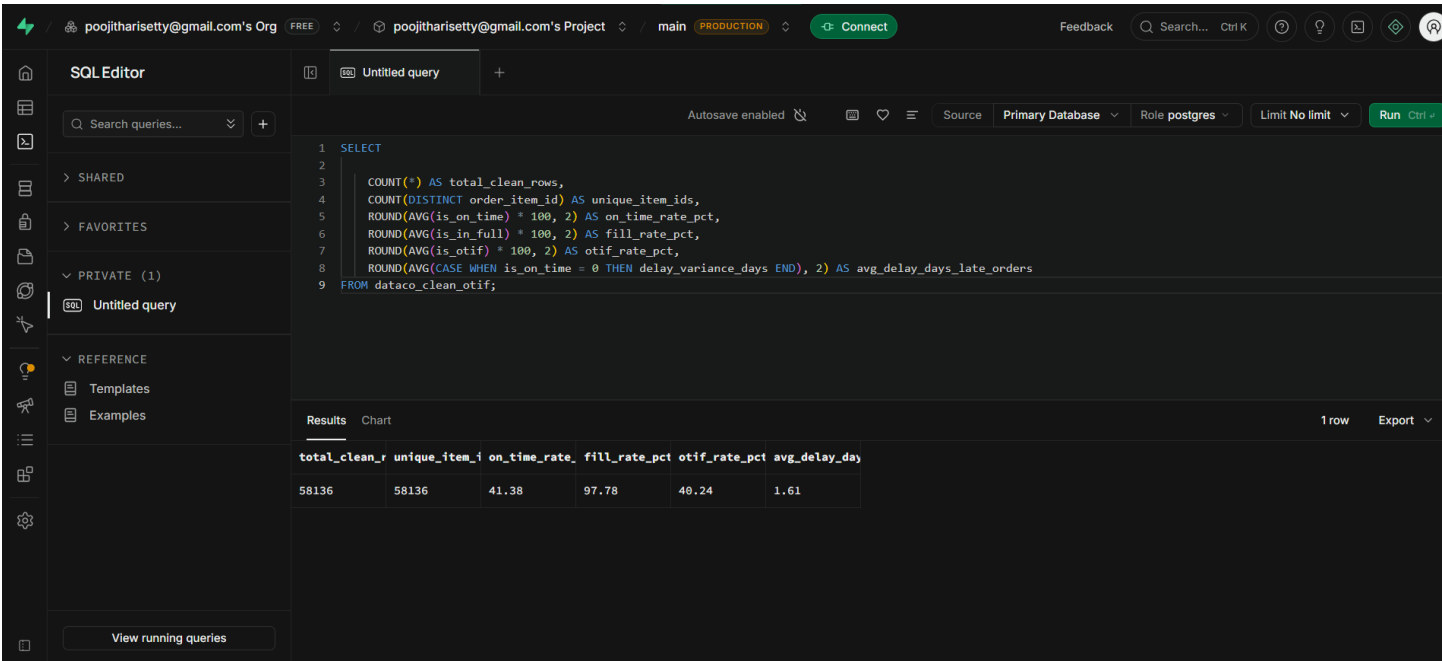
3. Exploratory Data Analysis & Technical Architecture

- **Data Cleaning & Exploration (Quadratic):** Ingested the raw dataset into Quadratic's interactive grid environment. Conducted initial data validation, schema standardization (converting headers to lowercase snake_case), and preliminary cell-level aggregations.

The screenshot displays the Quadratic data exploration interface. On the left, a sidebar shows the 'Data' tab with a 'Table row count' of 58,136. Below this, a 'Thought' section contains a SQL query: `clean_df = df.copy() Select core supply chain & logistics columns cols = ['Order Id', 'Order Item Id', 'Customer Id', 'Order Region', 'Order Country', 'Category Name', 'Shipping Mode', 'Delivery Status', 'Order Status', 'Days for shipping (real)', 'Days for shipment (scheduled)', 'order date (DateOrders)', 'shipping date (DateOrders)', 'Sales', 'Order Profit Per Order']`. The main area shows a data table with columns: Order Id, Order Item Id, Customer Id, Order Region, Order Country, Category Name, Shipping Mode, Delivery Status, Order Status, and Days for shipping (real). The table is filtered to show 25 rows. The status bar at the bottom indicates '1 message left on your Free plan, Upgrade now'.

Order Id	Order Item Id	Customer Id	Order Region	Order Country	Category Name	Shipping Mode	Delivery Status	Order Status	Days for shipping (real)
77202	180517	20755	Southeast Asia	Indonesia	Sporting Good	Standard Class Advance	shipp	COMPLETE	
75939	179254	19492	South Asia	India	Sporting Good	Standard Class	Late delivery	PENDING	
75938	179253	19491	South Asia	India	Sporting Good	Standard Class	Shipping on ti	CLOSED	
75937	179252	19490	Oceania	Australia	Sporting Good	Standard Class	Advance shipp	COMPLETE	
75936	179251	19489	Oceania	Australia	Sporting Good	Standard Class	Advance shipp	PENDING_PAY	
75935	179250	19488	Oceania	Australia	Sporting Good	Standard Class	Shipping canc	CANCELED	
75934	179249	19487	Eastern Asia	China	Sporting Good	First Class	Late delivery	COMPLETE	
75933	179248	19486	Eastern Asia	China	Sporting Good	First Class	Late delivery	PROCESSING	
75932	179247	19485	Eastern Asia	China	Sporting Good	Second Class	Late delivery	CLOSED	
75931	179246	19484	Eastern Asia	China	Sporting Good	First Class	Late delivery	CLOSED	
75930	179245	19483	Eastern Asia	Japan	Sporting Good	Second Class	Shipping canc	SUSPECTED_FR	
75929	179244	19482	Southeast Asia	Indonesia	Sporting Good	Second Class	Late delivery	PENDING	
75928	179243	19481	Southeast Asia	Indonesia	Sporting Good	Second Class	Late delivery	PENDING	
75927	179242	19480	South Asia	India	Sporting Good	First Class	Late delivery	COMPLETE	
75926	179241	19479	South Asia	India	Sporting Good	First Class	Late delivery	PROCESSING	
75925	179240	19478	South Asia	India	Sporting Good	First Class	Late delivery	COMPLETE	
75924	179239	19477	Eastern Asia	Corea del Sur	Sporting Good	Second Class	Late delivery	PENDING_PAY	
75923	179238	19476	South Asia	India	Sporting Good	First Class	Late delivery	CLOSED	
75922	179237	19475	South Asia	India	Sporting Good	First Class	Late delivery	COMPLETE	
75921	179236	19474	South Asia	India	Sporting Good	Same Day	Shipping on ti	PENDING_PAY	
75920	179235	19473	South Asia	India	Sporting Good	Same Day	Shipping on ti	PROCESSING	
75919	179234	19472	South Asia	India	Sporting Good	Standard Class	Late delivery	PENDING	
75918	179233	19471	Oceania	Australia	Sporting Good	Second Class	Late delivery	PENDING	

- **Database Warehousing (PostgreSQL / Supabase):** Loaded the cleaned dataset into a Supabase PostgreSQL instance (dataco_clean_otif table) to perform structured SQL queries and calculate business metrics.



- **Visual Analytics (Python / Google Colab):** Connected the database to Google Colab using pandas, sqlalchemy, and plotly.express to generate interactive visualizations and regional bottleneck heatmaps.



4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in Quadratic to answer key business questions:

1. Baseline Fulfillment Performance (OTIF & Overall Metrics)

Overall_Metrics									Jul 27, 2026
total_ord...	baseline_...	on_time_r...	in_full_rat...	avg_delay...	early_pct	exact_on_...	late_pct	total_reve...	
58136	40.24	41.38	97.78	1.61	22.78	18.59	58.62	6759132.6610	

Analytical Summary:

Out of 58,136 total orders evaluated, our global On-Time In-Full (OTIF) success rate is critically low at 40.24%. The data proves that inventory availability is not the issue, as the In-Full Rate reaches 97.78%. Instead, severe delivery delays (41.38% On-Time Rate) account for over 98% of all fulfillment failures, delaying shipments by an average of 1.61 days.

2. Shipping Mode SLA Compliance Analysis

Shipping_Mode_Analysis			Jul 27, 2026
shipping_...	otif_failur...	avg_transi...	
First Class	100	1	
Second Class	79.73	2.5	
Same Day	49.08	1	
Standard Class	41.06	1.51	

Analytical Summary:

Premium expedited services suffer from extreme SLA non-compliance. First Class shipping records a 100% failure rate due to unrealistic 1-day fulfillment schedule promises, while Second Class fails 79.73% of the time. Standard Class performs best with a 41.06% failure rate due to realistic schedule buffers.

3.Regional Bottlenecks & Revenue Exposure

Regional_Analysis			Jul 27, 2026
order_regi...	otif_rate_...	revenue_...	
West Asia	36.9	180164.54351	
East Africa	36.99	60591.101221	
South of USA	37.09	124176.21235	
South Asia	37.65	303367.09577	
Eastern Europ	37.85	123976.06242	

Analytical Summary:

Fulfillment performance drops below 38% in key emerging markets. South Asia represents the largest single financial bottleneck in this tier, with over \$303,000 in revenue at risk due to delayed deliveries.

4.High Risk Product Categories & Profitability Impact

Business Question: Which product lines experience the highest cancellations/fraud, and how does compliance impact profit margins?

Category_Cancel_Fraud	
Category ...	cancel_fra...
Women's Golf	6.08
Sporting Good	5.88
Boxing & MMA	5.06
Golf Balls	4.5
Water Sports	4.43

Profit_Margin_Co...	
complan...	avg_profit...
Non-Complan	10.87
OTIF Complian	11.2

Analytical Summary:

High-value sports categories like Women's Golf (6.08%) experience elevated cancellation and fraud rates. Furthermore, OTIF-compliant orders yield a higher profit margin (11.20% vs 10.87%), confirming that fulfillment failures directly erode profitability through reshipments and refunds.

5. Business Recommendations

Based on the quantitative insights derived from PostgreSQL data modeling in Supabase, Quadratic grid analysis, and interactive Plotly visual analytics, the following four data-driven strategies are recommended to improve global supply chain performance:

1. Recalibrate Shipping SLA Promises

- Operational Issue: First Class shipping records a 100% SLA failure rate (0.0% OTIF), and Second Class fails 79.7% of the time, despite packages arriving in 1 to 2.5 days.
- Root Cause: The fulfillment engine hardcodes an unrealistic 1-day delivery schedule for First Class orders, failing to account for 24+ hour warehouse handling times.
- Actionable Recommendation: Adjust the scheduled transit time parameter in the order management system from 1 day to 2 days for First Class orders.

This zero-cost system recalibration will instantly eliminate artificial SLA breaches and align customer delivery expectations with actual operational capabilities.

2. Establish Regional Fulfillment Centers in High-Risk Markets

- Operational Issue: Delayed shipments in Western Europe (\$1.2M) and Central America (\$1.1M) expose over \$2.3 Million in order value to cancellation and refund risks.
- Root Cause: Cross-border transit distance and localized customs bottlenecks extend delivery lead times in primary revenue markets.
- Actionable Recommendation: Partner with Third-Party Logistics (3PL) providers or establish localized distribution hubs within Western Europe and Central America. Pre-positioning high-velocity inventory closer to end consumers will drastically reduce international transit days and safeguard working capital.

3. Optimize Pre-Transit Warehouse Processing Velocity

- Operational Issue: Overall In-Full order completion is optimal at 97.78%, but On-Time delivery drops to 41.38%, driving an average delay of 1.61 days on late orders.
- Root Cause: Supply chain bottlenecks are caused by order dispatch delays rather than stockouts or inventory shortages.
- Actionable Recommendation: Shift logistics optimization priorities away from inventory procurement and toward warehouse picking, packing, and carrier handover velocity. Implementing automated batch picking and optimizing dispatch workflows to shave 12 hours off processing times will resolve the majority of delivery variance.

4. Implement Targeted Fraud and Cancellation Risk Controls

- Operational Issue: Product categories such as Women's Golf (6.08%), Sporting Goods (5.88%), and Boxing & MMA (5.06%) experience significantly elevated cancellation and suspected fraud rates.
- Root Cause: Inadequate payment verification protocols at checkout lead to post-order flagging and cancellation delays.
- Actionable Recommendation: Integrate automated address verification systems (AVS) and fraud-scoring algorithms specifically for high-risk product categories to catch fraudulent transactions prior to warehouse processing, protecting fill rates.

